

Biological Science

FAMOUS PERSONALITIES WHO USED APPLIED SCIENCE

- **Dr. Edward Jenner** – treatment of smallpox from cowpox
- **Louis Pasteur** – anthrax vaccine
- **Robert Koch** – anthrax and tuberculosis
- **Jonas Salk** – injection, polio, weakened viruses
- **Albert Sabin** – OPV, live vaccines
- **Alexander Fleming** – discovered penicillin from culture-grown mold "penicillium"
- **Paul Erlich** – early development of chemotherapy

BIOLOGY AS A NATURAL SCIENCE

Natural Science – deals with the study of nature. It is subdivided into the physical and biological sciences

A. Physical science – study of nonliving matter

1. Physics – study of matter, its properties, and its behavior
2. Chemistry – study of matter, its composition, and the changes that it undergoes
3. Meteorology – atmospheric conditions
4. Volcanology – volcanoes and their activities
5. Mineralogy – the study of minerals, their nature, properties, and distribution
6. Astronomy – the study of heavenly bodies
7. Geology – study of rocks and minerals

B. Biological science – science of life

1. Botany – the study of plants
2. Zoology – the study of animals

- **Aristotle** – father of Biology; studied the hearts and brains of animals; correlated the warming-cooling functions of the heart and brain to human feelings and emotions
- **Galen** – greatest authority on human anatomy; dissected animals; uses Barbary ape as a model
- **Andreas Vesalius** – anatomy should be based on direct observation
- **William Harvey** – physiologists; circulation of blood; true function of the heart and the blood vessels
- **Marcelo Malpighi** – dealt with blood by observing capillaries in the lungs of a frog
- **Eucharis Jansen** – invented the early microscope
- **Antoine Van Leeuwenhoek** – father of microbiology

- **Marie Francois Binchat** – discovered organisms were made of tissues
- **Rene Dutrochet and Robert Hooke** – discovered cell as the basis of tissues
- **Carolus Linnaeus** – binomial system for identifying and naming organisms
- **Jean Lamarck and Charles Darwin** – theory of evolution
- **Gregor Mendel, Thomas Hunt Morgan, Barbara McClintock** – theories on genetics

SUBDIVISIONS OF BIOLOGY IN VARIOUS FIELDS OF SPECIALIZATION

1. **Morphology** – study of structures and forms of organisms
2. **Anatomy** – study of the parts or structures of organisms
3. **Physiology** – study of normal functions of parts of an organisms
4. **Cytology** – study of structures and functions of cells
5. **Histology** – study of tissues
6. **Embryology** – study of growth and development of new organisms
7. **Ecology** – study of the environment and the interrelationship of organisms in it
8. **Taxonomy** – study of classification and naming of plants and animals
9. **Genetics** – study of heredity or how an organism's characteristics are transmitted from parents to offspring
10. **Evolution** – study of the origin and differentiation of different kinds of organisms
11. **Paleontology** – study of fossils of living things and their distribution in time
12. **Microbiology** – study of microorganisms
 - a. Bacteriology – study of bacteria
 - b. Virology – study of viruses
 - c. Protozoology – study of protozoa or one-celled animals
13. **Biochemistry** – study of chemistry of living things
14. **Biogeography** – study of distribution of plants and animals on earth



SUBDIVISIONS OF BIOLOGY ACCORDING TO SPECIFIC KIND OF ORGANISMS

1. Entomology – study of insects
2. Helminthology – study of worms
3. Ichthyology – study of fish
4. Ornithology – study of birds
5. Mammaology – study of mammals
6. Conchology – the study of shells
7. Anthropology – study of man
8. Parasitology – study of organisms that live and subsist on or in other living organisms

IMPORTANT TOOLS IN BIOLOGY

1. Optical instruments – make use of lenses
2. Nonoptical instruments – do not make use of lenses
 - Centrifuges and ultracentrifuges – used to separate the parts of the cell by spinning it in a solution and breaking it into several layers according to weight (with the heaviest part setting at the bottom)
 - Microdissection apparatuses or micromanipulators – attached to microscopes and are used to control the movement of a variety of tiny instruments that enable biologist to work on cells or small specimens
3. Microscopes – operate on two basic principles
 - a. Resolution – ability to make objects or specimens visible
 - b. Magnification power – ability to enlarge an object
 - Simple microscope – single lens; magnifying glass
 - Light microscope – uses light to produce enlarge view of an object
 - Compound microscope – uses two lenses; once lens produce and image that is further magnified by the second lens
 - Electron microscope – uses streams of electrons that bend in much the same way as beams of light are bent by glass lens
 - a. Transmission electron microscopes
 - b. Scanning electron microscopes
 - Phase-contrast microscope – no need for staining
 - Stereomicroscope or binocular microscope – used in studying the surface or external structure of specimens

COMMON LABORATORY TECHNIQUES IN BIOLOGY

1. Fixation – soaking of a specimen in a fixative, like formaldehyde after cutting it into small species
2. Mounting – placing of a specimen on a slide for investigation
3. Embedding – placing of a fixed specimen in wax or plastic so that it can be easily sliced

4. Sectioning – slicing of an embedded specimen into very thin cuts by a microtome
5. Staining – coloring of a specimen to bring out more details in a specific cell or tissue structures
6. Centrifugation – use of centrifuge to spin suspended materials in a liquid so as to collect these materials in fractions or layers
7. Microdissection – use of tiny instruments to perform various operations on living cells; done under a microscope
8. Chromotogrpahy – separation and analysis of complex mixtures; color comparison is usually its basis
9. Electrophoresis – separation of substances composed of particles with electric charges
10. Spectrophotometry – use of light to analyze samples to determine what a substance is and how much of it is present in a sample
11. Cell and tissue culture – technique used to maintain living cells or tissues in a tissue culture outside the body

BIOLOGICAL TECHNOLOGY

- Fluorescence in situ – used to visualize and map the genetic material in an individual cell; important for understanding different chromosomal abnormalities and some mutation in genes
- BT corn – one of the biotech crops being studied for success in the third world. Short for bacillus thuringiensis – it provides natural pest-killing toxin to new corn varieties
- IR 68098 – aside from development of methods to protect crops from pests, rice production program is also intensified through better agricultural technology and the use of good seed varieties; better rice developed at MMSU; best for rainfed lowlands, resistant to major pests
- Philippine sea cucumber – yields anticancer cells
- UPLB – Lectins – sugars with anticancer activity from the brown sea cucumber Holothuria sp. – therapeutic effect on HIV
- Tubli root powder – controls fish predators in freshwater ponds